Cognitive depletion from AI dependency

"I'm working on depletion of human cognition due to over dependency on ai; could you provide the exact direction of how to look at the problem and come up with real world solution?" *Generated by LeapSpace*

Depletion of Human Cognition Under AI Dependence: Quick Insights

Key Findings Table

Key finding	Population / setting	What the evidence shows
Short-term gains can come with long-term cognitive costs	General cognitive offloading, classrooms, workplaces, professional decision support	AI and external offloading often improve immediate efficiency or reduce effort, but later unaided recall, transfer, independent judgment, and skill retention can weaken when reliance becomes habitual [1] [2] [3] [4] [5] [6] [7].
Dependence is driven by metacognition, not just access to tools	Memory tasks, learning environments, AI decision support	People offload more when confidence is low, tasks feel hard, or cognitive load is high; overreliance also increases when trust in AI is miscalibrated [8] [9] [10] [11] [12] [13].
Automation bias is strongest under pressure	Medicine, transport, multitasking, knowledge work	Users accept wrong AI outputs more often when workload, uncertainty, time pressure, or multitasking reduce verification capacity [14] [15] [16] [17] [18] [19].
“Explainability” is not automatically protective	Clinical and general AI decision-support studies	Some explanations improve calibration, but others increase fluency, acceptance of wrong outputs, or cognitive overload; explanation quality matters more than explanation presence [20] [21] [22] [23] [24] [25].
Scaffolded AI protects learning better than direct-answer AI	Math, writing, engineering, tutoring	Socratic, hint-based, reflective, and metacognitive support tends to produce better delayed no-AI transfer than unrestricted direct-answer AI [6] [7] [26] [27] [28].

Key finding	Population / setting	What the evidence shows
Real-world mitigation should target workflow, not just user attitude	Education, medicine, enterprise use	Active verification prompts, prediction-before-AI, selective friction, audits, and human-responsibility workflows reduce blind acceptance with manageable productivity tradeoffs in at least some settings [29] [30] [31] [32] [33] [34].

Direct Answer

The best way to look at your problem is as a **human-AI self-regulation problem**: study where AI replaces memory, attention, reasoning, and verification, then test whether people still perform well when the AI is removed [11] [12] [35].

A strong project direction is a **4-layer framework**:

1. **Cognitive offloading**: what mental work is being outsourced—memory, planning, decomposition, recall, writing, diagnosis [1] [2] [35].
2. **Calibration failure**: when do users trust AI too much or too little, and can they tell correct from incorrect outputs [11] [20] [36]?
3. **Deskilling**: after repeated use, can they still do the task unaided [4] [5]?
4. **Workflow conditions**: which pressures amplify dependence—time pressure, workload, interruptions, high complexity, opaque interfaces [14] [19] [37]?

For **real-world solutions**, do not aim to "remove AI." Redesign use around:

- **prediction before AI** [29] [38]
- **verification prompts and forced justification** [30]
- **scaffolded/Socratic AI instead of direct-answer AI** [6] [7] [26]
- **selective friction at high-risk moments** [29] [32] [33]
- **periodic no-AI practice to detect skill atrophy** [4] [5]

If you want an exact research design, choose one domain and compare:

- **No AI**
- **Unrestricted AI**
- **Scaffolded/reflective AI**

Then measure:

- immediate performance,

- confidence/trust calibration,
- verification behavior,
- delayed no-AI performance after days or weeks [3] [28] [39].

Confidence level: High - The direction above is directly supported by repeated patterns across memory, decision-support, learning, and intervention studies [1] [4] [7] [16] [30].

Study Scope

This report synthesizes evidence spanning cognitive science, HCI, education, workplace studies, medicine, and governance, with emphasis on 2018-2025 empirical and review evidence, plus a few foundational earlier frameworks on offloading, trust, and automation bias [18] [19] [35] [40]. Methods represented include experiments, quasi-experiments, randomized trials, classroom longitudinal studies, field studies, and scoping/systematic reviews [3] [6] [31] [41].

Assumptions & Limitations

The strongest evidence is for **short-term performance and reliance effects**, while direct long-term evidence on durable cognitive depletion after sustained AI use remains comparatively sparse [42] [43] [44] [45] [46] [47]. Effects also vary by domain, expertise, task risk, and interface design, so there is no single universal anti-dependence intervention [7] [21] [48] [49].

Suggested Further Research

- Run **longitudinal workplace studies** measuring pre-, during-, and post-AI independent performance [44] [46].
- Test **unexpected AI withdrawal** to measure rebound costs, false recall, and confidence miscalibration in modern GenAI contexts [50] [51].
- Compare **direct-answer vs Socratic vs reflective AI** using the same delayed no-AI outcomes across domains [6] [7] [28].
- Study **heterogeneous users**: novices vs experts, high vs low need for cognition, high vs low workload contexts [30] [48].
- Evaluate whether guardrails still work **months after deployment**, not just in one-session experiments [52] [53] [54].

Full Report

1. Foundational Context

The central issue is not simply that AI makes work easier. The issue is that AI can shift cognition from **internal processing** to **external retrieval and acceptance**: people may remember where to find information rather than the information itself, rely on AI judgments instead of constructing their own, and move from active problem solving to passive verification [2] [35].

The main concepts needed to analyze the problem are:

- **Cognitive offloading**: using tools to reduce memory or reasoning demands [35].
- **Transactive/external memory**: storing knowledge in systems outside the self, often remembering location better than content [2].
- **Metacognitive regulation**: monitoring one's own confidence, difficulty, and need to verify [55] [56].
- **Automation bias**: accepting machine output despite contrary evidence or failing to detect its errors [19] [57] [58].
- **Trust calibration**: matching trust to actual system reliability rather than trusting globally [11] [59].
- **Cognitive load / deep processing tradeoff**: reducing immediate load can improve performance while weakening schema formation and later transfer [37] [60] [61].

Taken together, these concepts suggest that AI dependence should be studied as a tradeoff between **short-run efficiency** and **long-run cognitive autonomy** [1] [12].

Confidence level: High - These mechanisms recur across the evidence base and are explicitly named in foundational and applied studies [35] [37] [40].

2. Mechanisms: How AI Dependence Can Deplete Cognition

2.1 Cognitive offloading and memory externalization

A robust pattern is that offloading helps immediate performance but often reduces memory for offloaded content [1] [62] [63] [64]. In search-like contexts, people often retain location or retrieval routes better than the underlying information [2] [65]. When support is unexpectedly removed, recall worsens and false recall can increase [50] [66] [67].

This mechanism matters for AI because generative systems make offloading faster and broader: users can outsource summarization, synthesis, planning, drafting, and explanation, not just storage [68] [69] [70] [71].

2.2 Metacognitive shortcuts

Offloading is not random. People offload more when they feel less confident, when tasks are harder, and when effort feels high [8] [9] [72]. Higher cognitive load can lower confidence and judgments of learning, which likely increases later offloading choices [10] [73].

In AI settings, this means dependence can grow because the user learns a self-protective rule: “when uncertain, defer to the system” [12] [74]. That is adaptive in some situations, but harmful when it becomes default behavior rather than calibrated strategy [13] [75].

2.3 Reduced attention and shallow processing

Cognitive load theory implies that lowering immediate processing demands can also lower elaboration and schema construction if support replaces rather than scaffolds thinking [37] [60] [61]. Education studies report that AI can alter self-regulated learning sequences and may produce “metacognitive laziness,” even when short-term outputs improve [12] [13].

Some studies also report reduced cognitive effort, weaker creativity, and lower skill retention under sustained AI-assisted writing-tool use [76]. However, effects on measured cognitive load are mixed: ChatGPT and related tools do not consistently reduce load across tasks [77] [78].

2.4 Automation bias and trust miscalibration

In decision-support contexts, users often follow incorrect AI advice or fail to dismiss it [58] [79]. This risk rises under workload, multitasking, complexity, or uncertainty because verification becomes more demanding [14] [15] [17] [19].

Trust calibration studies show that explanation and confidence cues can improve judgment only when they help users discriminate correct from incorrect outputs; otherwise, they can increase acceptance without improving verification [11] [20] [23].

Confidence level: High - These mechanisms are supported across multiple subfields and converge conceptually [8] [11] [12] [16].

3. Observed Impacts in Real Tasks

3.1 Memory, recall, and retention

The clearest direct GenAI memory-cost evidence in the provided set is educational: using ChatGPT as a study aid was associated with worse delayed retention 45 days later than traditional study methods [3]. Earlier offloading evidence also shows poorer later recall and vulnerability when external support disappears unexpectedly [50] [66].

There is also evidence that external-store manipulation can induce false recall, especially when users trust the store contents [67].

3.2 Critical thinking and analytical reasoning

AI advice can reduce reflective reasoning when users accept output without scrutiny. In one study using the Cognitive Reflection Test, faulty AI support sharply reduced correct answers relative to a no-support control, while a warning nudge substantially improved performance [36]. Personnel-selection experiments also found participants failed to dismiss incorrect AI advice [79].

Taken together, these findings suggest that the main cognitive loss is not raw intelligence, but **reduced verification and reflective override** [80] [81].

3.3 Decision quality in high-stakes settings

Clinical evidence shows omission and commission errors under AI support, with one randomized study reporting omission errors of 15.9% and commission errors of 14.8% [16]. Incorrect AI recommendations were accepted uncritically in about 22.2% of decisions in one clinical decision-making study [58]. A chest radiography crossover study found many revised judgments changed correct calls into incorrect ones after AI review [82].

The same pattern appears beyond medicine, including teacher evaluation and psychology-related judgments [83] [84].

3.4 Deskilling and loss of independent competence

Direct long-term deskilling evidence is scarcer but important. A corporate-finance knowledge-work case reported that after the system had taken over most work and was later discontinued, workers could no longer complete the process unaided and had regressed to advanced-novice performance [4]. A multicenter colonoscopy trial found detection performance fell when endoscopists returned to non-AI procedures after repeated AI use [5].

These are among the strongest practical warnings that dependence can become visible only when support is withdrawn [4] [5].

3.5 Learning transfer and self-regulated learning

Not all AI support harms learning. Several controlled studies found positive transfer when AI was used as a tutor or scaffold rather than an answer engine [26] [27] [85]. Longitudinal classroom studies also found gains in self-regulated learning and engagement when AI support was structured [41] [86].

The important distinction is therefore **mode of assistance**: reflective/scaffolded designs support later independence more than unrestricted direct-answer use [7] [28].

Confidence level: Medium-High - Strong for immediate reliance and some delayed outcomes; lower for generalizing long-term deskilling across all domains because the direct longitudinal base is thinner [44] [47].

4. What Makes AI Use Harmful or Protective

4.1 Harm-amplifying conditions

Dependence risk increases with:

- high workload and multitasking [17] [18]
- time pressure and complexity [15] [19]
- opaque or overly fluent explanations [23] [24]
- low accountability for checking [87] [88]

4.2 User differences

Less experienced workers rely more on algorithmic scores, while greater expertise can reduce reliance [48] [89]. Need for cognition also matters: low-NFC users benefited especially from active verification prompts [30]. AI literacy can buffer uncritical acceptance, though trust effects are complex and not uniformly linear [90] [91] [92].

4.3 Interface and explanation design

Explanations help when they support discrimination between good and bad advice, such as nearest-neighbor examples or uncertainty-calibrated communication [11] [20]. But visible reasoning can also backfire and increase acceptance of wrong recommendations [23]. Moderate transparency may help, whereas excessive transparency can overload users and reduce effective use [21].

4.4 Why some AI support improves learning

Scaffolded AI works better than direct-answer AI because it preserves the learner's need to generate, justify, revise, and self-monitor [7] [26] [28]. By contrast, direct-answer systems can suppress productive struggle and deep processing [6] [12].

Confidence level: High - The protective vs harmful pattern appears consistently across education, HCI, and decision-support studies [6] [20] [30].

5. Real-World Solutions

5.1 A practical problem-framing model

Use this operational chain:

Task -> cognitive function outsourced -> reliance behavior -> immediate gain -> no-AI performance cost -> intervention point [35] [93] [94]

Example:

- drafting -> reasoning/writing outsourced -> accept AI prose -> faster output -> weaker independent writing transfer -> replace direct answer with scaffolded revision [28]

5.2 Five evidence-backed interventions

A. Prediction-before-AI workflow

Require the user to make an initial judgment before seeing AI output [29] [38]. This preserves cognitive ownership and reduces anchoring.

B. Active verification prompts

Prompt users to check evidence, compare alternatives, or justify accept/override decisions. This reduced overreliance and improved accuracy in controlled studies [30].

C. Scaffolded / Socratic AI

Use hints, questions, reflective revision loops, and delayed assistance rather than instant final answers [6] [7] [26]. This consistently outperforms unrestricted direct-answer AI for delayed no-AI performance.

D. Selective friction

Add pauses, structured reflection, or cognitive forcing only at high-risk moments rather than everywhere [29] [32] [33]. This can reduce blind acceptance, though some designs increase cognitive demand or reduce satisfaction [29].

E. AI literacy training

Teach how AI works, where it fails, what uncertainty means, and how to cross-check outputs [91] [92] [95]. The strongest supported content is foundational understanding plus structured practice, not generic awareness alone [91].

5.3 Organizational workflow design

In higher-stakes settings, keep AI in a role that supports rather than replaces human judgment:

- AI assists routine screening or drafting [96] [97]
- humans keep decomposition, exception handling, and final judgment [94] [98]
- disagreement triggers review or escalation [31] [99]
- deployments are monitored post hoc through audits and feedback loops [52] [53] [100]

This is the strongest practical direction in the literature: redesign roles so human agency remains active rather than nominal [93] [94].

5.4 A concrete research and implementation plan

For one domain, run a 3-condition study:

1. **No AI**
2. **Unrestricted AI**
3. **Scaffolded AI**

Measure:

- task accuracy and speed [101]
- confidence and trust calibration [11] [102]
- verification behavior / override rate [34] [102]
- delayed no-AI performance [3] [39]
- performance after unexpected withdrawal [50]
- periodic no-AI “cognitive resilience checks” to detect hidden deskilling early [4] [5]

This last step is partly an inference from the deskilling evidence: periodic no-AI testing is not established as a universal standard, but it follows directly from studies showing skill loss becomes visible when support is removed [4] [5].

Confidence level: Medium-High - Strong for the component interventions; somewhat lower for any one universal package, since effectiveness depends on task type and user context [34] [49].

6. Decision Framework: Exact Direction for Your Work

If your goal is to “look at the problem exactly,” the best framing is:

Research question

When does AI support shift from useful augmentation to harmful cognitive substitution? [1] [19]

Unit of analysis

Study one domain at a time:

- education,
- workplace knowledge work,
- high-stakes diagnosis,
- consumer everyday cognition [3] [4] [34]

Outcomes to track

- memory retention,
- independent reasoning,
- verification behavior,
- trust calibration,
- rebound after AI removal [5] [36] [50]

Intervention logic

Do not compare “AI vs no AI” only. Compare **unrestricted AI vs scaffolded AI vs no AI** [6] [28]. That design is more likely to reveal which assistance patterns preserve cognition while retaining performance [26] [30].

Practical success criterion

A good solution is not one that maximizes speed. It is one that preserves:

- productivity,
- independent task competence,
- accurate verification,
- calibrated trust [11] [31] [94]

Confidence level: High - This framework matches the most consistent cross-source pattern in the evidence [1] [7] [16] [94].

7. Conclusion

The evidence does support concern about cognitive depletion from overdependence on AI, but the most defensible interpretation is specific: AI tends to help immediate performance while risking weaker recall, poorer verification, deskilling, and reduced independent transfer when use becomes passive, uncalibrated, or routine under pressure [1] [4] [12] [16].

The best real-world solution is not prohibition. It is **human-AI workflow redesign**:

- think first,
- use AI second,
- justify acceptance,
- verify actively,
- practice periodically without AI,
- prefer scaffolded help over direct answers [7] [29] [30] [33].

That is the clearest evidence-backed direction for preserving cognition without giving up AI's practical benefits [94] [96] [97].

Overall confidence: Medium-High - The short-term mechanisms and intervention principles are well supported, but long-term cognitive adaptation evidence remains less complete than the strength of current claims sometimes implies [44] [47].

References

1. Skulmowski, A. (2023) The Cognitive Architecture of Digital Externalization. *Educational Psychology Review*. https://doi.org/10.1007/s10648-023-09818-1
2. Sparrow, B., Liu, J., Wegner, D.M. (2011) Google effects on memory: Cognitive consequences of having information at our fingertips. *Science*. https://doi.org/10.1126/science.1207745
3. Barcauí, A. (2025) ChatGPT as a cognitive crutch: Evidence from a randomized controlled trial on knowledge retention. *Social Sciences and Humanities Open*. https://doi.org/10.1016/j.ssaho.2025.102287
4. Sutton, S.G., Arnold, V., Holt, M. (2023) An extension of the theory of technology dominance: Capturing the underlying causal complexity. *International Journal of Accounting Information Systems*. https://doi.org/10.1016/j.accinf.2023.100626
5. Heudel, P.E., Crochet, H., Filori, Q., (...), Blay, J.Y. (2026) Artificial intelligence in medicine: a scoping review of the risk of deskilling and loss of expertise among physicians. *ESMO Real World Data and Digital Oncology*. https://doi.org/10.1016/j.esmorw.2026.100693
6. Bassner, P., Lenk-Ostendorf, B., Beinstingel, R., (...), Krusche, S. (2026) Less stress, better scores, same learning: The dissociation of performance and learning in AI-supported programming education. *Computers and Education: Artificial Intelligence*. https://doi.org/10.1016/j.caeai.2025.100537
7. Ates, H. (2026) Human-centered GenAI feedback design in higher education: a multisite experiment on direct, reflective, and hybrid approaches to scientific argumentation. *International Journal of Educational Technology in Higher Education*. https://doi.org/10.1186/s41239-026-00614-9
8. Boldt, Annika, Gilbert, Sam J. (2019) Confidence guides spontaneous cognitive offloading. *Cognitive Research: Principles and Implications*. https://doi.org/10.1186/s41235-019-0195-y
9. Gilbert, S.J. (2024) Cognitive offloading is value-based decision making: Modelling cognitive effort and the expected value of memory. *Cognition*. https://doi.org/10.1016/j.cognition.2024.105783
10. Conte, N., Fairfield, B., Padulo, C., Pelegrina, S. (2023) Metacognition in working memory: Confidence judgments during an n-back task. *Consciousness and Cognition*. https://doi.org/10.1016/j.concog.2023.103522
11. Steyvers, M., Tejeda, H., Kumar, A., (...), Smyth, P. (2025) What large language models know and what people think they know. *Nature Machine Intelligence*. https://doi.org/10.1038/s42256-024-00976-7

12. Efklides, A. (2011) Interactions of metacognition with motivation and affect in self-regulated learning: The MASRL model. *Educational Psychologist*. https://doi.org/10.1080/00461520.2011.538645

13. Fan, Y., Tang, L., Le, H., (...), Gašević, D. (2025) Beware of metacognitive laziness: Effects of generative artificial intelligence on learning motivation, processes, and performance. *British Journal of Educational Technology*. https://doi.org/10.1111/bjet.13544

14. Holm, Søren, Stanton, Catherine, Bartlett, Benjamin (2021) A New Argument for No-Fault Compensation in Health Care: The Introduction of Artificial Intelligence Systems. *Health Care Analysis*. https://doi.org/10.1007/s10728-021-00430-4

15. https://www.sciencedirect.com/science/article/pii/S1471227X26001734

16. Mirkin, Sophia, Albensi, Benedict C. (2023) Should artificial intelligence be used in conjunction with Neuroimaging in the diagnosis of Alzheimer's disease?. *Frontiers in Aging Neuroscience*. https://doi.org/10.3389/fnagi.2023.1094233

17. Alami, J., El Iskandarani, M., Riggs, S.L. (2025) The Effect of Workload and Task Priority on Multitasking Performance and Reliance on Level 1 Explainable AI (XAI) Use. *Human Factors*. https://doi.org/10.1177/00187208251323478

18. Parasuraman, R., Manzey, D.H. (2010) Complacency and bias in human use of automation: An attentional integration. *Human Factors*. https://doi.org/10.1177/0018720810376055

19. Goddard, K., Roudsari, A., Wyatt, J.C. (2012) Automation bias: A systematic review of frequency, effect mediators, and mitigators. *Journal of the American Medical Informatics Association*. https://doi.org/10.1136/amiajnl-2011-000089

20. Humer, C., Hinterreiter, A., Leichtmann, B., (...), Streit, M. (2024) Reassuring, Misleading, Debunking: Comparing Effects of XAI Methods on Human Decisions. *ACM Transactions on Interactive Intelligent Systems*. https://doi.org/10.1145/3665647

21. Ngo, V.M. (2025) The AI transparency dilemma: when more is less for trust and adoption. *Information Discovery and Delivery*. https://doi.org/10.1108/IDD-03-2025-0056

22. Leichtmann, B., Humer, C., Hinterreiter, A., (...), Mara, M. (2023) Effects of Explainable Artificial Intelligence on trust and human behavior in a high-risk decision task. *Computers in Human Behavior*. https://doi.org/10.1016/j.chb.2022.107539

23. Zhu, Z., Li, Y. (2026) Over-reliance on AI in information tasks: Unraveling the roles of reasoning visibility and task complexity. *Telematics and Informatics*. https://doi.org/10.1016/j.tele.2026.102464

24. Cau, F.M., Spano, L.D. (2025) The Influence of Curiosity Traits and On-Demand Explanations in AI-Assisted Decision-Making. *International Conference on Intelligent User Interfaces, Proceedings IUI*. https://doi.org/10.1145/3708359.3712165

25. Kuhl, U., Bush, A. (2026) When Bias Backfires: The Modulatory Role of Counterfactual Explanations on the Adoption of Algorithmic Bias in XAI-Supported Human Decision-Making. *Communications in Computer and Information Science*. https://doi.org/10.1007/978-3-032-08333-3_12

26. ter Sarkisian, A.G., Miller, C., Poonjolai, N., (...), Atweh, J. (2026) Albert or Alberta? Investigating Socratic Dialogue and Modality in AI-Assisted Engineering Problem Solving. *2026 Systems and Information Engineering Design Symposium, SIEDS 2026*. https://doi.org/10.1109/SIEDS69358.2026.11540271
27. Kerimbayev, N., Menlibay, Z., Shadiev, R., (...), Adamova, K. (2025) The influence of Generative AI on the transfer of algorithmic skills in students. *International Conference Automatics and Informatics, ICAI 2025 - Proceedings*. https://doi.org/10.1109/ICAI67591.2025.11324568
28. Huang, S.Y.B. (2026) Designing a scaffolded GenAI revision environment for EMI academic writing: a randomized active-control study of no-ai transfer and feedback use. *Interactive Learning Environments*. https://doi.org/10.1080/10494820.2026.2713874
29. https://www.sciencedirect.com/science/article/pii/S1435565525005074
30. Doh, W., Goh, Y., Kim, S.-H. (2026) Forcing Deliberation: The Role of Cognitive Interventions in Preventing Overreliance in AI-Assisted Decision-Making. *Lecture Notes in Computer Science*. https://doi.org/10.1007/978-3-032-29548-4_13
31. Santiago, R.L.V., Lins, I.D. (2026) AI-powered innovation tournament evaluations: A hybrid NLP approach. *Technovation*. https://doi.org/10.1016/j.technovation.2026.103622
32. Zhang, C., Ju, K., Bidoshi, P., (...), Rzeszotarski, J.M. (2025) Friction: Deciphering Writing Feedback into Writing Revisions through LLM-Assisted Reflection. *Conference on Human Factors in Computing Systems - Proceedings*. https://doi.org/10.1145/3706598.3714316
33. Pe, S., Bergomi, L., Nicora, G., (...), Parimbelli, E. (2026) Cognitive Friction in Clinical Decision Support: A Comparative Study of Judicial and Adjunct Human–AI Interaction Protocols. *Machine Learning and Knowledge Extraction*. https://doi.org/10.3390/make8070216
34. Cabitza, F., Campagner, A., Tontini, G.E. (2026) Calibrating Reliance: Addressing Misuse and Disuse in AI-Based Second-Opinion Systems for Medical Diagnosis. *Proceedings of the AAAI Conference on Artificial Intelligence*. https://doi.org/10.1609/aaai.v40i47.41457
35. Risko, E.F., Gilbert, S.J. (2016) Cognitive Offloading. *Trends in Cognitive Sciences*. https://doi.org/10.1016/j.tics.2016.07.002
36. Wingerter, T.L., Straub, T., Schweitzer, S. (2025) Mitigating Automation Bias in Generative AI Through Nudges: A Cognitive Reflection Test Study. *Procedia Computer Science*. https://doi.org/10.1016/j.procs.2025.09.331
37. Sweller, J. (1988) Cognitive load during problem solving: Effects on learning. *Cognitive Science*. https://doi.org/10.1016/0364-0213(88)90023-7
38. Wang, Y., Yu, H., Zhang, L., (...), Luo, S. (2026) The role of explainable AI in influencing user trust and advice-taking in image aesthetic quality evaluation. *International Journal of Human Computer Studies*. https://doi.org/10.1016/j.ijhcs.2026.103798

39. Aldamen, Hesham, Alawneh, Salwa Mohammad, Almashour, Mohamad, (...), Alsharefeen, Rami (2026) Climate-control ChatGPT for EFL writing and AI literacy. *Frontiers in Artificial Intelligence*. https://doi.org/10.3389/frai.2026.1869343

40. Lee, J.D., See, K.A. (2004) Trust in automation: Designing for appropriate reliance. *Human Factors*. https://doi.org/10.1518/hfes.46.1.50_30392

41. Zhang, J., Derakhshan, A. (2025) Integrating 'GPT-4o' as an AI Tool Into K-12 Contexts: Assessing Its Long-Term Effects on Students' Self-Regulated Learning (SRL) and Task Engagement Through Latent Growth Curve Modelling. *European Journal of Education*. https://doi.org/10.1111/ejed.70329

42. Yuh, M., Byeon, S., Jain, N., Hwang, I. (2022) Evaluation of Cognitive State Feedback for Accelerating Human Learning. *Proceedings of the American Control Conference*. https://doi.org/10.23919/ACC53348.2022.9867342

43. Bartlett, M.L., McCarley, J.S. (2024) Experts and Non-Experts Use Diagnostic Aids Inefficiently. *Proceedings of the Human Factors and Ergonomics Society*. https://doi.org/10.1177/10711813241275516

44. Paglialunga, Andrea, Melogno, Sergio (2025) The Effectiveness of Artificial Intelligence-Based Interventions for Students with Learning Disabilities: A Systematic Review. *Brain Sciences*. https://doi.org/10.3390/brainsci15080806

45. Lawrence, R., Dodsworth, E., Massou, E., (...), Fulop, N.J. (2025) Artificial intelligence for diagnostics in radiology practice: a rapid systematic scoping review. *eClinicalMedicine*. https://doi.org/10.1016/j.eclinm.2025.103228

46. Hautz, W.E., Marcin, T., Schauber, S.K., (...), Kämmer, J.E. (2026) Self-Monitoring Accuracy over Time for a Complex Diagnostic Skill. *Perspectives on Medical Education*. https://doi.org/10.5334/pme.1767

47. Li, Wei, Xu, Ya, Yao, Le, Liu, Yantong (2026) Exploring the influence of AI self-study rooms on K–12 learners' motivation, self-regulation, enjoyment, and engagement. *Frontiers in Psychology*. https://doi.org/10.3389/fpsyg.2026.1768389

48. Fischer-Abaigar, U., Kern, C., Barda, N., Kreuter, F. (2024) Bridging the gap: Towards an expanded toolkit for AI-driven decision-making in the public sector. *Government Information Quarterly*. https://doi.org/10.1016/j.giq.2024.101976

49. Shibuya, Y., Amichai-Hamburger, Y. (2026) AI disclosure formats and user responses to AI-generated video: Evidence from a cross-national experiment. *Computers in Human Behavior*. https://doi.org/10.1016/j.chb.2026.109113

50. https://www.sciencedirect.com/science/article/pii/S1069938426001178

51. Bonham, A.J., González-Vallejo, C. (2009) Assessment of calibration for reconstructed eye-witness memories. *Acta Psychologica*. https://doi.org/10.1016/j.actpsy.2009.02.008

52. https://www.sciencedirect.com/science/article/pii/S1573689X26001258

53. Tang, S., Zhu, H. (2024) Mitigating bias in generative AI: a comprehensive framework for governance and accountability. *Law, Ethics and Technology*. https://doi.org/10.55092/let20240008

54. Krasteva-Hristova, R. (2026) Professional Judgment and AI Disclosure Governance in Audit and Sustainability Assurance: Public Evidence from the UK Big Four. *Journal of Risk and Financial Management*. https://doi.org/10.3390/jrfm19090675

55. Flavell, J.H. (1979) Metacognition and cognitive monitoring: A new area of cognitive-developmental inquiry. *American Psychologist*. https://doi.org/10.1037/0003-066X.34.10.906

56. de Bruin, A.B.H., Roelle, J., Carpenter, S.K., (...), Waldeyer, J. (2020) Synthesizing Cognitive Load and Self-regulation Theory: a Theoretical Framework and Research Agenda. *Educational Psychology Review*. https://doi.org/10.1007/s10648-020-09576-4

57. Lyell, David, Magrabi, Farah, Raban, Magdalena Z., (...), Coiera, Enrico (2017) Automation bias in electronic prescribing. *BMC Medical Informatics and Decision Making*. https://doi.org/10.1186/s12911-017-0425-5

58. Kucking, F., Busch, D.A., Przysucha, M., (...), Hübner, U. (2026) Impact of AI recommendation correctness on diagnostic accuracy in clinical decision-making. *International Journal of Medical Informatics*. https://doi.org/10.1016/j.ijmedinf.2025.106223

59. Yu, Y., Gomez-Cabello, C.A., Haider, S.A., (...), Forte, A.J. (2025) Enhancing Clinician Trust in AI Diagnostics: A Dynamic Framework for Confidence Calibration and Transparency. *Diagnostics*. https://doi.org/10.3390/diagnostics15172204

60. Sweller, J., Van Merrienboer, J.J.G., Paas, F.G.W.C. (1998) Cognitive Architecture and Instructional Design. *Educational Psychology Review*. https://doi.org/10.1023/A:1022193728205

61. Leppink, J., Paas, F., Van Der Vleuten, C.P.M., (...), Van Merrienboer, J.J.G. (2013) Development of an instrument for measuring different types of cognitive load. *Behavior Research Methods*. https://doi.org/10.3758/s13428-013-0334-1

62. Morrison, A.B., Richmond, L.L. (2020) Offloading items from memory: individual differences in cognitive offloading in a short-term memory task. *Cognitive Research: Principles and Implications*. https://doi.org/10.1186/s41235-019-0201-4

63. Arreola, H., Flores, A.N., Latham, A., (...), Vu, K.-P.L. (2019) Does the Use of Tablets Lead to More Information Being Recorded and Better Recall in Short-Term Memory Tasks?. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. https://doi.org/10.1007/978-3-030-22660-2_20

64. Grinschgl, S., Neubauer, A.C. (2022) Supporting Cognition With Modern Technology: Distributed Cognition Today and in an AI-Enhanced Future. *Frontiers in Artificial Intelligence*. https://doi.org/10.3389/frai.2022.908261

65. Alghamdi, O., Clinch, S., Skeva, R., Jay, C. (2023) How are websites used during development and what are the implications for the coding process?. *Journal of Systems and Software*. https://doi.org/10.1016/j.jss.2023.111803

66. Richmond, Lauren L., Kearley, Julia, Schwartz, Shawn T., Hargis, Mary B. (2023) Take a load off: examining partial and complete cognitive offloading of medication information. *Cognitive Research: Principles and Implications*. https://doi.org/10.1186/s41235-023-00468-z

67. Risko, Evan F., Kelly, Megan O. (2025) Offloading memory: A review. *Learning and Memory: A Comprehensive Reference*. https://doi.org/10.1016/B978-0-443-15754-7.00014-6

68. Wang, P., Liu, H., Zou, L., Paas, F. (2026) Overloaded minds and machines: a cognitive load framework for human-AI symbiosis. *Artificial Intelligence Review*. https://doi.org/10.1007/s10462-026-11510-z

69. Tanveer, M. (2026) Artificial intelligence literacy moderating the effects of artificial intelligence use on cognitive outcomes including memory retention decision making and cognitive laziness. *Discover Psychology*. https://doi.org/10.1007/s44202-026-00697-5

70. Su, J., Wang, Z., Qiang, Z., Wang, Z. (2026) Is It Mine or Not Mine? Cognitive Offloading to Generative AI and Concern over the Erosion of Cognitive Ownership. *Journal of Intelligence*. https://doi.org/10.3390/jintelligence14080182

71. García-Barrios, D.A. (2025) Epistemic Partner or Cognitive Crutch: A Conceptual Model of Cognitive Offloading in Human–Artificial Intelligence Collaboration. *IEEE Internet Computing*. https://doi.org/10.1109/MIC.2025.3626694

72. Dreisbach, Gesine, Jurczyk, Vanessa (2021) The role of objective and subjective effort costs in voluntary task choice. *Psychological Research*. https://doi.org/10.1007/s00426-021-01587-2

73. Wang, T., Li, S., Tan, C., (...), Lajoie, S.P. (2023) etacognitive aCognitive load patterns affect temporal dynamics of self-regulated learning behaviors, metacogntive judgments, and learning achievements. *Computers and Education*. https://doi.org/10.1016/j.compedu.2023.104924

74. Kulal, A. (2025) Cognitive Risks of AI: Literacy, Trust, and Critical Thinking. *Journal of Computer Information Systems*. https://doi.org/10.1080/08874417.2025.2582050

75. Klingbeil, A., Grützner, C., Schreck, P. (2024) Trust and reliance on AI — An experimental study on the extent and costs of overreliance on AI. *Computers in Human Behavior*. https://doi.org/10.1016/j.chb.2024.108352

76. Budiyono, H., Marzuki, Pudjaningsih, W., (...), Maulidina, A. (2025) Exploring Long-Term Impact of AI Writing Tools on Independent Writing Skills: A Case Study of Indonesian Language Education Students. *International Journal of Information and Education Technology*. https://doi.org/10.18178/ijiet.2025.15.5.2306

77. Schulz, T., Knierim, M.T. (2024) Cognitive Load Dynamics in Generative AI-Assistance: A NeuroIS Study. *45th International Conference on Information Systems, ICIS 2024*. https://www.scopus.com/pages/publications/105010827052

78. Avello-Martínez, R., Gajderowicz, T., Gómez-Rodríguez, V.G. (2024) Is ChatGPT helpful for graduate students in acquiring knowledge about digital storytelling and reducing their cognitive load? An experiment.. *Revista de Educación a Distancia*. https://doi.org/10.6018/red.604621

79. Cecil, J., Lermer, E., Hudeček, M.F.C., (...), Gaube, S. (2024) Explainability does not mitigate the negative impact of incorrect AI advice in a personnel selection task. *Scientific Reports*. https://doi.org/10.1038/s41598-024-60220-5

80. Xie, Xiao, Zhang, Lawrence Jun, Wilson, Aaron J. (2025) Comparing ChatGPT Feedback and Peer Feedback in Shaping Students' Evaluative Judgement of Statistical Analysis: A Case Study. *Behavioral Sciences*. https://doi.org/10.3390/bs15070884

81. Bani-Hani, I., Hujran, O., Anwar, M. (2026) The integration of AI with human judgement and critical thinking in decision-making process: a focus on design phase. *Journal of Decision Systems*. https://doi.org/10.1080/12460125.2026.2652347

82. Lemke, T., Marka, A.W., Prucker, P., (...), Busch, F. (2026) Artificial Intelligence-Assisted Chest Radiography: A Prospective Crossover Multi-Reader Study on Diagnostic Performance and Workflow Efficiency. *Academic Radiology*. https://doi.org/10.1016/j.acra.2026.08.014

83. van Zyl, Llewellyn E. (2026) The unintended negative consequences of artificial intelligence use for psychologists. *Frontiers in Psychology*. https://doi.org/10.3389/fpsyg.2026.1729050

84. Du, Peitao, Liu, Tingting, Xian, Xujin (2026) Automation bias in teachers' evaluation of student writing: effects of algorithmic warnings and visual risk cues in AI detection reports. *Frontiers in Psychology*. https://doi.org/10.3389/fpsyg.2026.1889402

85. Syromiatnykov, Y., Mamatov, F., Chuyanov, K., (...), Egamov, M. (2026) Independent Engineering Transfer After Traceable Generative-AI-Assisted Learning: A Six-University Controlled Trial with Deterministic Cluster Allocation in Agricultural Engineering Education. *Applied Sciences (Switzerland)*. https://doi.org/10.3390/app16188949

86. Li, W., Xu, Y., Yao, L., Liu, Y. (2026) Exploring the influence of AI self-study rooms on K–12 learners' motivation, self-regulation, enjoyment, and engagement. *Frontiers in Psychology*. https://doi.org/10.3389/fpsyg.2026.1768389

87. McGuirl, J.M., Sarter, N.B. (2006) Supporting trust calibration and the effective use of decision aids by presenting dynamic system confidence information. *Human Factors*. https://doi.org/10.1518/001872006779166334

88. Jayabal, R., Ganesan, G., Tripathy, S., (...), Priya K, K. (2026) Artificial intelligence-enabled process safety: Advances, challenges, and future directions for high-risk industrial systems. *Journal of Loss Prevention in the Process Industries*. https://doi.org/10.1016/j.jlp.2026.106026

89. Logg, J.M., Minson, J.A., Moore, D.A. (2019) Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational Behavior and Human Decision Processes*. https://doi.org/10.1016/j.obhdp.2018.12.005

90. Huang, K.-T., Ball, C. (2024) The Influence of AI Literacy on User's Trust in AI in Practical Scenarios: A Digital Divide Pilot Study. *Proceedings of the Association for Information Science and Technology*. https://doi.org/10.1002/pra2.1146

91. Markus, A., Pfister, J., Carolus, A., (...), Wienrich, C. (2024) Effects of AI understanding-training on AI literacy, usage, self-determined interactions, and anthropomorphization with voice assistants. *Computers and Education Open*. https://doi.org/10.1016/j.caeo.2024.100176

92. Song, D., Zhang, P., Zhu, Y., (...), Zhou, L. (2025) Effects of generative artificial intelligence on higher-order thinking skills and artificial intelligence literacy in nursing undergraduates: A quasi-experimental study. *Nurse Education in Practice*. https://doi.org/10.1016/j.nepr.2025.104549

93. Alberdi, E., Povyakalo, A., Strigini, L., Ayton, P. (2004) Effects of incorrect computer-aided detection (CAD) output on human decision-making in mammography. *Academic Radiology*. https://doi.org/10.1016/j.acra.2004.05.012

94. Díaz-Rodríguez, N., Del-Ser-Lorente, J., Coeckelbergh, M., (...), Herrera, F. (2023) Connecting the dots in trustworthy Artificial Intelligence: From AI principles, ethics, and key requirements to responsible AI systems and regulation. *Information Fusion*. https://doi.org/10.1016/j.inffus.2023.101896

95. Kim, S.-Y. (2026) Developing pre-service English teachers' critical and ethical AI literacy in writing with generative AI. *System*. https://doi.org/10.1016/j.system.2026.104037

96. Pacheco, M., Rodrigues, R. (2026) Human–AI Collaboration in UX: Rethinking Creativity and Productivity in Understanding User Needs. *Interaction Design and Architecture(s)*. https://doi.org/10.55612/S-5002-068-007

97. Soomro, S., Fan, M., Soomro, S., (...), Shaikh, S.N. (2025) Balancing AI and human interaction: Impact of AI interaction on well-being. *AI in Mental Health: Innovations, Challenges, and Collaborative Pathways*. https://doi.org/10.4018/979-8-3373-5072-1.ch007

98. Herrera, F. (2025) Intelligent organizational engineering driven by human-AI collaboration and explainable Al to increase productivity. *Direccion y Organizacion*. https://doi.org/10.37610/87.702

99. Kang, X., Ding, W., Matsumoto, K., (...), Shi, X. (2027) Neuro-symbolic information fusion for explainable retrieval-augmented generation in mental health applications: A survey and challenges. *Information Fusion*. https://doi.org/10.1016/j.inffus.2026.104702

100. Hyytiäinen, T., Luukka, P. (2026) Explainable AI for Financial Cost Variance Analysis: A Human-in-the-Loop Workflow for Anomaly Screening and Explanation. *International Journal of Human-Computer Interaction*. https://doi.org/10.1080/10447318.2026.2721889

101. Vasquez, Holland M., Pianarosa, Emilie, Sirbu, Renee, (...), Rosella, Laura C. (2024) Human factors methods in the design of digital decision support systems for population health: a scoping review. *BMC Public Health*. https://doi.org/10.1186/s12889-024-19968-8

102. Yu, Yunguo, Gomez-Cabello, Cesar A., Haider, Syed Ali, (...), Forte, Antonio J. (2025) Enhancing Clinician Trust in AI Diagnostics: A Dynamic Framework for Confidence Calibration and Transparency. *Diagnostics*. https://doi.org/10.3390/diagnostics15172204